

Meta-Labeling To Trade or Not to Trade

Side, trade and size — three decisions, three stages



Meta-labeling splits a trade into three decisions

Each stage answers one question and is judged by one measure

STAGE 1 · SIDE DECISION

Long or short?

- Any rule or model
- Output: side ± 1

Recall

STAGE 2 · TRADE DECISION

Act on it or pass?

- A classifier
- Output: $p = P(\text{win})$

Precision

STAGE 3 · BET SIZE

How much?

- Kelly from p
- Output: bet size f^*

Kelly

Stage 1 catches the chances, stage 2 drops the losers, stage 3 sizes the rest.

Two measures, two stages: recall first, then precision

Positive = stage 1 says 'trade' · TP = winning trade · FP = losing trade · FN = missed winner

Recall — stage 1 maximises it

$$\text{Recall} = \frac{TP}{TP + FN}$$

A MISSED WINNER (FN)

Lost for good

no later stage can add it back

Precision — stage 2 maximises it

$$\text{Precision} = \frac{TP}{TP + FP}$$

A LOSING TRADE (FP)

Can still be dropped

stage 2 passes on it

That is why the order is fixed: recall first, precision second.

Stage 1 picks the side and misses as few chances as possible

This model already exists on most desks — a technical rule, a fundamental view or an ML model

WHAT IT DOES

- E.g. fast MA above slow MA → long
- Otherwise → short
- Side only — no size yet

HOW TO TUNE IT

- Loosen the trigger: more signals
- High recall, low precision is fine
- Stage 2 removes the losers

Output of stage 1: a side for every signal

$$\hat{s}_i \in \{-1, +1\}$$

The side is fixed here. No later stage changes it.

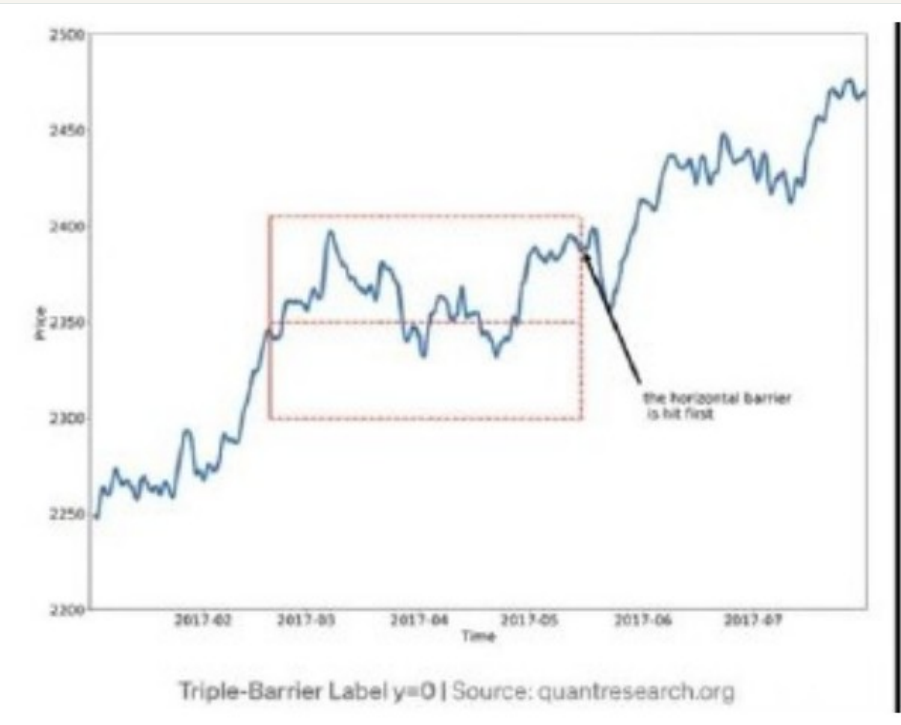
Stage 2 first needs to know which signals were right

From each signal, follow the price until it touches one of three barriers

THREE BARRIERS

- Profit-take: side won
- Stop-loss: side lost
- Time limit: sign of return

Barrier widths scale with volatility



The meta label is 1 only if the side made money

$\hat{s}$ = side from stage 1 · r = return until the first barrier touched · illustration with $\pm 2\%$ barriers

Meta label: what stage 2 learns to predict

$$z_i = 1 \quad \text{if} \quad \hat{s}_i r_i > 0, \quad z_i = 0 \quad \text{otherwise}$$

Signal	Side (stage 1)	First barrier touched	P&L of the side	Meta label z
#1	Long	Profit-take	+2.0%	1
#2	Long	Stop-loss	-2.0%	0
#3	Short	Profit-take	+2.0%	1
#4	Short	Time limit	-0.4%	0

Stage 2 trades only the signals likely to win

A binary classifier learns z from what stage 1 does not see — volatility, regime, signal strength

Output of stage 2: probability that this signal wins

$$p = P(z = 1 \mid \text{stage 1 fired, } X)$$

p HIGH

Trade

act on stage 1's side

p LOW

Pass

a likely false positive

EFFECT

Precision rises

fewer losing trades

Stage 2 never flips the side — it only says trade or pass.

Stage 3 sizes the bet with the Kelly criterion

p = win probability from stage 2 • $q = 1 - p$ • β = profit / loss, set by the barrier widths

Kelly fraction

$$f^* = \frac{\beta p - q}{\beta}$$

Symmetric barriers ($\beta = 1$)

$$\beta = 1: f^* = p - q = 2p - 1$$

READING IT ($\beta = 1$)

- $p = 0.5 \rightarrow f^* = 0$, pass
- $p = 0.6 \rightarrow$ bet 20%
- $p = 0.7 \rightarrow$ bet 40%

p must be calibrated: $p = 0.6$ should win 60% of the time.

Position = side from stage 1 × Kelly fraction from stage 2's p .

Result: same signals, fewer losing trades, better risk

Trading example, out of sample 2018-01-04 to 2019-01-28

	Stage 1 alone	With meta-labeling
Cumulative return	19.7%	39.6%
Sharpe ratio	0.65	0.82
Max drawdown	−61.9%	−36.8%

Same side on every trade — only the trade decision changed.

STAGE 1 · SIDE: RECALL.

STAGE 2 · TRADE: PRECISION.

STAGE 3 · SIZE: KELLY.

**Catch the chances, drop the losers,
size the rest.**